MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025
THIRD SEMESTER
DISCIPLINE SPECIFIC ELECTIVE (DSE)-MG3DSEBOT200-ETHNOBOTANY AND INTELLECTUAL PROPERTY RIGHTS
(2024 ADMISSION ONWARDS)

Duration 2 hours Maximum marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Part A

Very short answer

Answer any **10** questions

Each question carries **2** marks

1. Define ethnobotany. [K]/ [CO1]

Ethnobotany is the scientific study of the interrelationship between people and plants. It focuses on how indigenous and local communities use plants for their cultural, religious, medicinal, nutritional, and economic needs.

2. Give any two contributions of EK Jankiammal. [K]/ [CO1]

Pioneering Indian botanist and cytogeneticist; Conducted research on medicinal and economically important plants of India; Advocated for the documentation and conservation of traditional plant knowledge in tribal and rural communities; Played a crucial role in indigenous plant classification systems and promoted the importance of native flora in Indian science; Was associated with the Botanical Survey of India, and influenced policies on plant conservation and ethnomedicine. **(any two points)**

3. Which plant flowers are used by Malampandaram in forest deity worship? [K]/ [CO1]

Any flower used for rituals **(Liberal valuation)**

4. Name the medicinal plant producing 'Ajmalin'. [K]/ [CO1]

Rauwolfia serpentina

5. Write short note on medicinal value of 'Aswagandha'. [K]/ [CO1]

A well-known Rasayana in Ayurveda; Used to treat Fatigue and weakness, Stress and anxiety, Joint pain and arthritis, Male infertility and low libido, Immune deficiency, neurodegenerative diseases etc; Contains withanolides –with anti-inflammatory and anti-cancer properties ; Helps in reducing cortisol, improving memory, and increasing muscle mass; Used as a nutraceutical and herbal supplement. (**Any two points**).

6. What is the purpose of analysing logbook of forest in ethnobotanical research? [K]/ [CO5]

Log book of forests help to find out sale volumes, frequency, species harvested etc.

7. What are the benefits of plant variety registration for breeders? [K]/ [CO6]

Breeder gets the following rights on registering a variety -Right to produce, sell, market, distribute, import, or export the protected variety; Can give license or permission to others to use the variety; Can take legal action against infringers (unauthorized use); Variety is protected for 15 years (crops) and 18 years (trees & vines). (**Any two points**)

8. What type of information is documented in PBR? [K]/ [CO3]

People's Biodiversity Register (PBR) document local biological resources including species, cultivars, and landraces and the associated traditional knowledge and conservation practices.

9. Expand IPO. [K]/ [CO6]

Indian Patent Office / Intellectual Property Organization

10. What is the full form of TKDL in IPR? [K]/ [CO8]

Traditional Knowledge Digital Library

11. What is the purpose of National gene bank? [K]/ [CO5]

Preserve genetic diversity of plants and crops; Protect rare and endangered plant species; Provide seeds for future research, breeding, and restoration; Act as a backup in case of natural disasters, climate change, or crop failure. **(Any two)**

12. Define the term 'patent'. **[K]/ [CO6]**

Patent is the exclusive right granted by the Government to the owner of the invention to make, use, sell, manufacture, import his patented invention and prevent others from making, using, selling, manufacturing, importing the invention, without his permission.

Part B

Short answer

Answer any **8** questions

Each question carries **5** marks

13. Define Ethnobotany as an interdisciplinary science. **[K]/ [CO1]**

Ethnobotany is an interdisciplinary science which combine insights from the following fields:

- Botany-Identification classification of plants.
- Anthropology- Study of cultural practices and social context.
- Pharmacology-Validation of medicinal properties.
- Ecology-Understanding plant-environment relationships.
- Agriculture-Traditional farming and crop management.
- Lingustics-study of local plant names.
- Sociology-analysis of knowledge transmission.
- Economics- Understanding plant-based livelihoods.
- Law & IPR-Protection of traditional knowledge and rights.

This integration makes ethnobotany a complete science.

14. How is *Pongamia pinnata* useful in agriculture? **[U]/ [CO1]**

Improves soil fertility through nitrogen fixation; parts are used for animal feed, natural pesticides, and green manure; valuable for soil restoration and agroforestry; the oil cake is a rich source of nutrients for both fertilizer and animal feed. **(Liberal valuation)**

15. Recall the Kurichya festival that involves plant-based rituals. **[K]/ [CO1]**

The Kurichya festivals that involve significant plant-based rituals are the Thira festival (Malakkari Thira) and festivals related to their agricultural cycle, such as Thulapathu and Puthari. **(Liberal valuation)**

16. Describe the role of *Trichopus zeylanicus* in the traditional health practices of Kani tribe. **[U]/ [CO4]**

Trichopus zeylanicus is known as Arogyapacha; Used by Kani tribes for boosting energy and stamina, reducing fatigue during long forest treks, Improving immunity and vitality; serve as a potent health tonic for enhancing energy, vitality, and immunity; The tribe consumes the unripe fruits and leaves to remain energetic; believe that regular consumption keeps them youthful, strong, and resistant to various illnesses.

17. **Discuss the role of ethnobotany in promoting sustainable development and conservation of biodiversity**. **[U]/ [CO5]**

Biodiversity Conservation-Indigenous knowledge helps identify rare, endangered, and useful species; Encourages community-led conservation efforts like sacred groves and herbal gardens.

Promote Sustainability-Traditional practices offer adaptive strategies for farming, forestry, and water conservation; Promotes resilience through biodiversity-based livelihoods; encourages sustainable harvest of plant resources.

18. Explain the theoretical frameworks that underlie the concept of benefit sharing of wealth in ethnobotany. **[A]/ [CO3]**

Benefit sharing refers to the fair and reasonable distribution of benefits arising from the use of traditional knowledge and biological resources held by local or indigenous communities. Eg. Jeevani -explanation (Link between TBGRI and Kani tribes). (**3 marks**)

Encourages mutual benefit through conservation incentives, community participation, and sustainable utilization of plant resources. Views biological resources and traditional knowledge as the collective heritage of humankind. Emphasizes equity, justice, and fairness in sharing benefits between resource providers; the use of biodiversity should ensure long-term ecological, economic, and social sustainability. (**Any two-2 marks**)

19. Name any 5 GI tagged products from Kerala and their socioeconomic value. [U]/ [CO7]

GI-Tagged Products from Kerala

Kodungallur Pottuvellari, Nilambur teak, Vazhakkulam pine apple, Aramula kannadi, Marayur Jaggery, Attappady Thuvara, Attappady Aattukombu, Onattukara Ellu, Kanthaloor-Vattavada Veluthulli, Attappady Aattukombu Amara, Kuthampully Dhoties & Set Mundu, Chendamangalam Dhoties & Set Mundu, Payyanur Pavithra Ring, Kuthampully Sarees, Kannadippaya, Pokkali rice, Wayanad Gandhakasala rice. Wayanad jeerakasala rice etc (**Any 5**)

Socioeconomic value - Enhances marketability and premium pricing for local products.; Protects traditional knowledge and craftsmanship; Guards against unauthorized use of regional names and products ; enhance their competitiveness in domestic and international markets

20. Differentiate between process patent and product patent. [U]/ [CO8]

A product patent protects a final invention, such as a drug or device, regardless of how it was made. Eg A patent on a new chemical compound or a specific medication. A process patent protects the specific method or technique used to create a product, but not the product itself. Eg A patent on the novel method used to synthesize that same chemical compound. **Liberal valuation.**

21. Discuss the importance of community participation in safeguarding traditional knowledge. [U]/ [CO8]

Community participation ensures preservation of traditional knowledge (TK); promotes transmission of TK between generations; recognizes the knowledge holders as the primary owners; ensures that the knowledge is documented accurately.

22. How do patents contribute to industrial and economic development. [K]/ [CO6]

Patents provide inventors with exclusive rights to their inventions for a limited time, protecting them from direct competition; promote the sharing of technical knowledge; attract investment; encourage them to invest in R&D; leads to the creation of new products and services, which contribute to economic development; The patent's disclosure requirement makes the technology publicly available for others to build upon, accelerating further innovation.

Part C

Essay

Answer any 1 question

Each question carries 10 marks

23. Write an account of the ethnobotanical practices involving *Coscinium fenestratum*, *Dioscorea*, *Vitex negundo* and *Gloriosa superba*. [An]/ [CO4]

Coscinium fenestratum-

The stem and root are rich in berberine, used in traditional medicine; Used for treating fever, diabetes, jaundice, ulcers, and skin diseases; Tribal communities use the yellow dye extracted from the stem for colouring fabrics and for ritualistic purposes; Antimicrobial and anti-inflammatory properties recognized in Ayurveda and Siddha systems.

Dioscorea spp

Edible tubers are a staple food for many tribal groups during food scarcity; Used for treating rheumatism, constipation, and stomach disorders; Certain species (like *D. oppositifolia*, *D. alata*) are rich in diosgenin, a steroidal compound used in modern pharmaceutical industries; Tribal women use some Dioscorea species for contraceptive and menstrual regulation.

Vitex negundo

Leaves are used as a poultice for joint pain, swelling, and wounds; Smoke from burning leaves is used as a mosquito repellent in rural households; Decoction of leaves used to treat fever, respiratory issues, and menstrual disorders; Considered a sacred plant in some communities and planted near homes**.**

Gloriosa superba

Rhizomes and seeds are used in small quantities for treating snake bites, gout, and ulcers; Highly toxic if consumed in large quantities; used carefully by experienced tribal healers; Also used in fertility-related rituals and traditional abortifacient remedies; Contains colchicine, a compound used in modern medicine for treating gout and cancer research.

24. Explain the step by step procedure for obtaining patent. [A]/ [CO6]

Step 1: Filing of Patent Application- To be filed at the appropriate branch of the Patent Office under whose jurisdiction he normally resides. Applicant shall submit the necessary documents while filing the application. The patent applications can also be filed electronically through the website of the patent office (www.ipindia.nic.in). Upon receiving the patent application, Patent Office will issue the patent application number and date of filing, to the applicant.

Step 2: Publication-The patent application will be published in the official journal of Patents after the expiry of 18 months from the date of filing or date of priority of the application.

Step 3: Opposition-Upon publication of the patent application, any person can file pre grant opposition to the Controller of Patents against the grant of patent.

Step 4: Request for Examination/Expedited Examination

The applicant or any interested person must file request for examination for getting their patent application examined by the Patent Office. Every patent application will be examined only up on the receipt of such request.

Step 5: Issuance of First Examination Report (FER) -The Patent Office will conduct detailed examination on the patent application to check whether the invention possesses the criteria of novelty. inventiveness and industrial application and issues a First Examination Report (FER) to the applicant, which contains the objections raised by the Patent Office.

Step 6: Reply to the Examination Report

The applicant must give his/her response to the objections and clarifications in the First Examination Report within 6 months from the date of issue of the FER. If the applicant is not able the response within the prescribed time, the application will be abandoned.

Step 7: Grant of Patent and issuance of Certificate

The applicant meets the objections and clarifications asked for in the FER, the Controller will grant patent and the details of the Patent will be entered in the register of Patents. Patent Certificate will be issued to the applicant within one month from the date of grant of patent

Step 8: Renewal fees

The duration of the Patent is 20 years from the data of filing of the application. To keep the patent in force, renewal fees should be paid before the expiration of the second year or the succeeding year from the date of fling of patent.

Step 9: Post grant opposition

Upon grant of patent also, any interested person, based on different grounds, may file a post grant opposition [Section 25(2)] in Form 7 along with prescribed fee to the Controller against the grant of patent, within one year from the date of publication of grant of patent.